18. Friday, May 9. Back To Set Theory (5.1, 5.2)

Previously: (Lectures 4 and 5)

- A **set** is a well-defined collection of objects.
- For subsets A, B of some universal set, we defined
  $A \cup B$, $A \cap B$, $A \setminus B$, A^c.

Definition 1: Let A be a subset of a universal set U.
The **power set of A**, denoted $\mathcal{P}(A)$ is the set of all subsets of A, that is:

$$\mathcal{P}(A) = \{X \subseteq U \mid X \subseteq A\}$$

Note: · The elements of $\mathcal{P}(A)$ are themselves sets.
· We know that $\emptyset$ and A are subsets of A, so $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.

Example: · For $A = \emptyset$, the only subset of $\emptyset$ is $\emptyset$. Thus $\mathcal{P}(\emptyset) = \{\emptyset\}$ 1 element

If $A = \{a, b\}$, then $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ 4 elements

· If $A = \{a, b, c\}$, then $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$ 8 elements.

Definition: The **cardinality** of a finite set is the number of elements in that set. Notation: $\text{Card}(A)$ or $|A|$.

Exercise: Prove that, for every integer $n \geq 0$, if $\text{card}(A) = n$ then $\text{Card}(\mathcal{P}(A)) = 2^n$.

Sketch: Base case: $\text{card}(\mathcal{P}(\emptyset)) = 1 = 2^0$

Inductive step: Take A with $k+1$ elements. Fix the last element of A, call it α.

Any subset X of A satisfies exactly one of the following:

1) $\alpha \notin X$.
2) $\alpha \in X$.

By Ind Hyp, there are 2^k subsets satisfying (1). Why?

There is the same amount satisfying (2). Why?

Proving facts about sets (5.2)

Recall: For subsets A, B of some universal set U, we defined $A \subseteq B$ to mean "for every $x \in U$, if $x \in A$ then $x \in B$."

Technique (Element-Chasing). To prove $A \subseteq B$, take an arbitrary element $x \in A$, and show that $x \in B$.
To disprove $A \subseteq B$, find an example of some $x \in A$ such that $x \notin B$.

Note: "arbitrary $x \in A$" means that you can't choose a specific element $x \in A$.

Theorem 2: Let $A = \{n \in \mathbb{Z} \mid n \text{ is a multiple of } 4\}$ and $B = \{n \in \mathbb{Z} \mid n \text{ is even}\}$. Then $A \subseteq B$.

Non-proof: Since $4 = 4 \cdot 1$, we see that $4 \in A$. Also, since $4 = 2 \cdot 2$, we see $4 \in B$. Hence $A \subseteq B$.

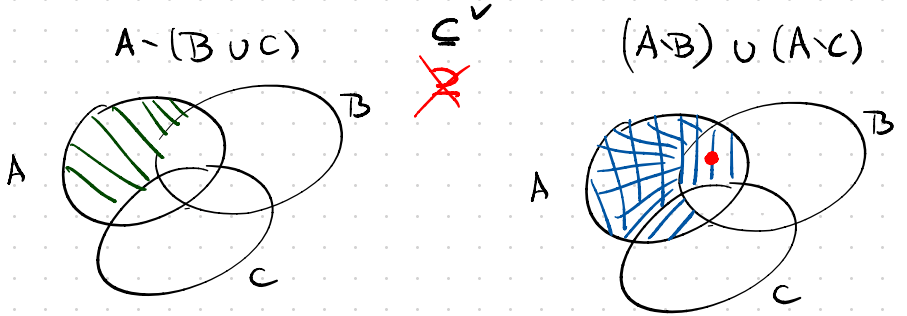
Proof: Let $n \in A$. Since n is a multiple of 4, there exists $k \in \mathbb{Z}$ such that $n = 4 \cdot k$. But then $n = 4 \cdot k = 2 \cdot 2k$, and by closure of $\mathbb{Z}$, we have $2k \in \mathbb{Z}$. Hence $x = 2(2k)$ is even, so $n \in B$. We conclude that $A \subseteq B$. □

Theorem 3: Let A, B, C be sets. Then $A \setminus (B \cup C) \subseteq (A \setminus B) \cup (A \setminus C)$

Proof: Let $x \in A \setminus (B \cup C)$. By definition, this means $x \in A$ and $x \notin B \cup C$.
Thus $x \notin B$ and $x \notin C$.
Since $x \in A$ and $x \notin B$, we have $x \in A \setminus B$, so also $x \in (A \setminus B) \cup (A \setminus C)$

Therefore $A \setminus (B \cup C) \subseteq (A \setminus B) \cup (A \setminus C)$ ∎

Question: What about the reverse inclusion?



The reverse inclusion doesn't hold in general. Something in $(A \cap B) \setminus C$

Example: Take $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{2, 3\}$

Then $(A \setminus B) \cup (A \setminus C) = \{2\} \cup \{1\} = \{1, 2\}$

But $A \setminus (B \cup C) = \{1, 2\} \setminus \{1, 2, 3\} = \emptyset$

So $(A \setminus B) \cup (A \setminus C) \not\subseteq A \setminus (B \cup C)$

Technique: To prove that two sets A, B are equal, prove $A \subseteq B$ and $B \subseteq A$.

Theorem 4: Let A, B, C be sets. Then $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

Proof: We will prove both inclusions

[$\subseteq$] Suppose $x \in A \setminus (B \cup C)$. Then by definition, $x \in A$ and $x \notin B \cup C$. Since $x \notin B \cup C$ we have $x \notin B$ and $x \notin C$. Therefore $x \in A \setminus B$ and $x \in A \setminus C$, Thus by definition $x \in (A \setminus B) \cap (A \setminus C)$. This shows $\subseteq$.

[$\supseteq$] Suppose $x \in (A \setminus B) \cap (A \setminus C)$. Since $x \in (A \setminus B)$, this means $x \in A$ and $x \notin C$.

Similarly, $x \in A$ and $x \notin C$. Since $x \notin B$ and $x \notin C$, we have $x \notin B \cup C$.

Therefore $x \in A \setminus (B \cup C)$ This shows $\supseteq$.

Since we have proved both inclusions, the desired equality follows $\square$

Definition 5: Let A and B be sets. We say that A and B are disjoint if $A \cap B = \emptyset$.

Theorem 6: Let A and B be sets. If $A \subseteq B$, then A and B^c are disjoint.

Proof: It suffices to prove the contrapositive. Suppose A and B^c are not disjoint. Then there exists $x \in A \cap B^c$. Therefore $x \in A$ and $x \in B^c$. In other words, $x \in A$ and $x \notin B$, so $A \not\subseteq B$.

□